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June 18, 2026

To the editors of *PLOS ONE*:

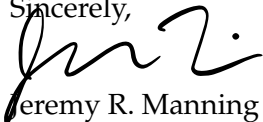
I have enclosed our manuscript entitled *A Stylometric Application of Large Language Models* to be considered for publication as a *Research Article*. The manuscript shows that large language models can be used to distinguish the writings of different authors: an individual GPT-2 model, trained from scratch on the works of a single author, predicts that author's held-out text more accurately than text written by others.

Our central claim is supported by a systematic and fully reproducible set of experiments. We trained individual models on the works of eight classic authors and used cross-entropy loss as a measure of stylistic similarity, an approach we term predictive comparison. To confirm that our findings are robust rather than artifacts of a single training run, we trained models across multiple random seeds and several text representations (full text, content words, function words, and part-of-speech sequences). These ablations show that both content words and function words contribute to author-specific signatures, whereas grammatical structure alone is less distinctive. We further apply the approach to a real attribution problem, supporting R. P. Thompson's authorship of the well-studied 15th book of the *Oz* series, originally attributed to L. F. Baum.

We designed the study with reproducibility as a priority, in keeping with the standards of *PLOS ONE*: all code and data needed to reproduce every result, figure, and statistical test are openly available at <https://github.com/ContextLab/llm-stylometry>. We believe the work will interest the journal's broad, interdisciplinary readership, including researchers in computational linguistics, digital humanities, and machine learning.

This manuscript reports original primary research that is not under consideration for publication elsewhere. An earlier version has been posted as a preprint on arXiv (arXiv:2510.21958), consistent with *PLOS ONE*'s preprint policy. Thank you for considering our manuscript; I hope you will find it suitable for publication in *PLOS ONE*.

Sincerely,



Jeremy R. Manning